

Element I

Tests:

<i>Design Requirement</i>	<i>Test</i>	<i>Explanation</i>
<i>Changes color manually and automatically</i>	Number of Color Options: 13 Color Options Output Speed: <0.1 sec	<i>The 13 color options are displayed on the remote controller for the fiber optic wave board. The colors manually selected on the controller will display on the board. To calculate the number of color options we counted the number of colored buttons on the remote. To calculate output speed we used a stopwatch. The clock was started when an input was selected, then stopped when the result was visible. This process was repeated 10 times to assure accuracy.</i>
<i>Provides calming effect</i>	Speed of light Changing Jump 7: 0.8 sec Jump 3: 1.6 Fade 3: 1.6 Fade 7: 0.8 Flash: Instant	<i>The group used a stopwatch to find the time for one cycle of the fade setting, and then divided by the number of colors the fade changed between to find the rate at which the colors fade and change.</i>
<i>Illuminates the sensory room</i>	Light range: Highest brightness: 6ft. Lowest brightness: 29in.	<i>To calculate this the group used a light measuring app called Light Meter LM-3000. The group held the light meter in front of the fiber optic wave board on its highest and lowest brightness settings. Respectively, one member of the group would slowly move the light meter away from the wave board until the exact point the light</i>

		<i>meter ceased to record a light measurement to find the range of the light and illuminating capacity of the board.</i>
<i>Brightness can be altered</i>	Light Output (Standard Brightness): 1.3 Lux Lowest brightness: 1 Lux Highest brightness: 11 Lux Brightness options: 9	<i>To calculate this, the group used a light measuring app called Light Meter LM-3000. The group placed the fiber optic wave board in a dark environment, ensuring any outside sources of light would not impact the Lux measurement. One member of the group held the Light Meter directly in front of the fiber optic light display, and varied the brightness from lowest to highest, recording the measured Lux for each brightness.</i>
Integrated into the room	Ask Ms. Brennan for a visual assessment	<i>The group will ask Ms. Brennan and the other teachers to provide a visual assessment of the fiber optic wave board and how well it integrates into the rest of the room. The group will also ask the teachers to measure the students' opinions about the board.</i>